

Composing parametric memory: what BDH's uniform neuron axis makes possible

Concept: Parametric Memory in LLMs · Team Invariance, IIT Kharagpur · DataForge 2026, Pathway track · Artifact: bdh-merger.vercel.app

The design pressure. A Transformer splits memory in two: fixed weights hold what it learned in training, a growing key–value cache holds what it just read. Neither composes well. Combining two fine-tuned Transformers means averaging their weights (model soups, arXiv:2203.05482) or adding task vectors (arXiv:2212.04089) — both force two specialists into one parameter set, and interference is the normal outcome, with a literature devoted to mitigating it (TIES-Merging, arXiv:2306.01708). The question underneath is where memory lives and whether trained capability can be **composed** rather than retrained.

What BDH changes. The Dragon Hatchling (BDH, arXiv:2509.26507) is a post-Transformer architecture that reformulates attention as synaptic memory: a population of n neurons whose pairwise connections strengthen Hebbianly as the model reads, so reasoning and short-term memory share one computational fabric instead of living in weights and a cache respectively. The structural consequence this submission exploits is that every learnable parameter hangs off one uniform neuron axis. In the reference implementation the model is five matrices — three of shape $n \times d$ (encoder, encoder_v, decoder) and two of shape $\Omega \times d$ (embedding, Im_head, Ω = vocabulary size) — giving **$3nd + 2\Omega d$** parameters. Attention contributes no learnable parameters, and layers reuse the same matrices.

The operation that unlocks. Because neurons are interchangeable units on a single axis, two BDH models can be **concatenated**: stack the three $n \times d$ matrices ($n_A + n_B$ neurons) and average the two $\Omega \times d$ matrices. This is the merge rule BDH's paper sketches in §7.1. It is neither weight averaging nor ensembling — no parameter is blended away, and inference is a single forward pass, not two. A Transformer admits no equivalent, because its parameters are not organised on one shared, permutation-tolerant axis.

Approach	Operation	Params	Inference	Requires
Ensembling	run both, combine outputs	2×	2 passes	nothing
Weight averaging	mean of weights	1×	1 pass	shared init
Task arithmetic	add task vectors	1×	1 pass	shared init
BDH concatenation	stack the neuron axis	2×	1 pass	uniform neuron axis

Concatenation trades parameter count for keeping both specialists intact at single-pass inference cost — a trade only the uniform axis makes available.

What we tested, and what failed. §7.1 raises composability as a conjecture — “when the model latent space promotes concept disentangling then it is feasible to directly compose concepts” — but does not test it. We did, on toy-scale reimplementations ($n = 1024$, $\sim 209k$ parameters per parent) trained on paired synthetic languages sharing a pivot vocabulary, with θ controlling vocabulary overlap: $11 \theta \times 3 \text{ seeds} \times 3 \text{ datasets}$. Concatenation beat averaging in **33 of 33 runs on all three datasets, at every θ** , and merge damage tracks θ closely ($\text{corr} = -0.991$ over per- θ means). But **our own hypothesis was falsified**: we predicted damage would track measured representational overlap M (Hungarian-matched neuron activation correlation) and would localise to identifiable “collision” neurons. M stayed flat and its correlation with damage flipped sign under re-aggregation; ablating top-collision neurons never beat a size-matched random control, and was worse at $k = 100$. The disentanglement-implies-composability claim we were testing is the Transformer-side finding of arXiv:2305.12827; we could not confirm its BDH analogue.

Evidence labels. Everything above is developer-reported by us, on a labelled toy reimplementation — not an official BDH model, not an independent reproduction, not a deployment. BDH's own §7.1 merge result is likewise reported by the architecture's authors. We are aware of no external reproduction of BDH merging. The artifact runs real forward passes in the browser (JS port verified against PyTorch to $3.5e^{-7}$); precomputed sweeps are labelled as such in the UI.

Roles of BDH and BDH-CQ. BDH is central: the merge operation is a direct consequence of its parameterisation. **BDH-CQ has no direct role in this concept and we claim none** — it concerns in-context skill acquisition without weight updates, a different mechanism from composing already-trained parameters.

The most important limitation. We cannot yet separate “representational overlap does not predict mergeability” from “our metric is mis-specified.” M pools over the whole neuron axis, while the divergence we suspect matters sits in the output pathway. Cross-dataset testing sharpened this rather than settling it: at 40 concepts M stopped being flat and correlated with damage at -0.935 , so M 's behaviour is sensitive to vocabulary size rather than being a fixed property of the architecture. A parameter-count control — merged $2n$ against a single $2n$ parent, which would rule out a pure size effect — was planned and never run.

Where to continue. Primary sources: BDH (arXiv:2509.26507, §7.1 for the merge rule); model soups (2203.05482); task arithmetic (2212.04089); TIES-Merging (2306.01708); weight disentanglement (2305.12827). Full numbers, the retracted claims, and every limitation are in the repository README.